

JUHI DAMLEY

jdamley28@cmc.edu | (305) 538-2856 | San Francisco, CA

[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

EDUCATION

Claremont McKenna College
Bachelor of Arts, Computer Science

Claremont, CA
August 2024 - May 2028

Harvey Mudd College

Claremont, CA

Major and coursework is done at HMC through the cross-campus major program

Activities: Girls Who Code (Founder and President), Sagehen Capital Management, Circle K, Museum of Art Club, Board of Trustees Advancement Committee, The Feedbacker, Women in Math, HMC Grader and Tutor for CS5, p-AI

Relevant coursework: Abstract Algebra, Stochastic Operations Research, Linear Algebra, Statistics, Discrete Math, Calculus III, Intro to Game Theory, Data Structures and Program Development, Intermediate Microeconomics

EXPERIENCE

Financial Economics Institute | Research Analyst

January 2026 - Present

- Implemented Python portfolio-optimization algorithms for the "Optimized Indexation" project, including cost- and constraint-aware routines scalable to large asset universes
- Built a config-driven Python pipeline that reproducibly ingests CRSP-style panels, derives deterministic run keys and per-stage seeds, and checkpoints typed JSON + Parquet artifacts for full auditability
- Developed statistical kernels (sample means, unbiased covariances) with NaN-safe windowing and numerical-stability checks, enabling reliable moment estimation on irregular financial panels
- Utilized HPC resources to process and analyze large datasets and optimized code for performance and scalability

The Gould Center for Humanistic Studies | Research Assistant

October 2024 - Present

- Developed professional website to showcase research, publications, courses, and related content using TypeScript, Vercel, Vite, Node, and React
- Delivered election research for Prof. Gaston Espinosa's "Religion, Race, Gender, and the 2024 Election" series spanning conferences, a five-book series, and upcoming podcast
- Managed marketing assets, social outreach, and task delegation for multi-speaker conference
- Analyzed datasets and shipped 70+ charts/visuals for presentations and publications
- Assessed scholarly viability of potential research projects by webscraping data and querying OpenAI API

Claremont McKenna College | Office of Advancement Intern

October 2024 - December 2025

- Strategically advised Board of Trustees on AI-integration opportunities and threats for Advancement
- Managed alumni data pipeline: integrated third-party platforms, digitized files, and standardized records
- Developed Python program to partially automate data updates, cutting processing time 50%+
- Cultivated custom Gemini model to automate classification of 4000+ alumnae

SKILLS AND INTERESTS

Programming Languages: Python, Java, Racket, R, C++, TypeScript

AI/Product: : Prompt Engineering, LLM Integration, Figma, Notion, Slack

Tools / Platforms: Git, LaTeX, CRM, Adobe Photoshop, Canva, Excel, Word, Powerpoint, Vercel, Firebase, Node.js, Supabase

Interests: Knitting and Crochet, Vinyl Records, Books, The Sims, Cats, Gaming, Urban Hiking, Ghibli, Concerts

PROJECTS

- LectureTeX** | Site Python — An end-to-end lecture-to-PDF pipeline served via FastAPI with React web and Electron desktop clients that uses, speech-to-text via Faster Whisper on Modal A100 GPUs, subject-aware LaTeX note generation across 10 LLMs, and automated post-processing and compilation.
- Prioritize** | Site TypeScript — Priority-queue web app that helps users manage and organize tasks.
- WakeUp** | Swift — Accessibility-focused iOS and watchOS application designed to detect excessive daytime sleepiness via Apple Watch sensors and provide vibrations to induce wakefulness
- juhi.studio** | Site TypeScript — Portfolio featuring game of life animations, a live Mandelbrot fractal background, and a retro Windows 95 UI, integrated with two sub-applications.
- IE Employment Analysis** | GitHub R — Modeled and compared regional employment trends in the Inland Empire and Imperial County, developing "Job Opportunity Index" metric.
- Stylometry Model** | GitHub Python — Text-classification pipelines to compare Murakami novel translations.

HONORS & AWARDS

Handley Leadership Fellowship • National Merit Scholar • Venture Capital Program Finalist • CMC SP '25 Statistics Competition Winner • AHS Merit Scholarship (Approx. \$85,000)